TECHNOLOGICAL UNIVERSITY (THANLYIN)
DEPARTMENT OF INFORMATION TECHNOLOGY ENGINEERING

CLASSIFICATION AND PESTICIDE RECOMMENDATION FOR MAIZE LEAF DISEASES BASED ON VGG16

BY

MA PHYO PA PA LWIN

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GRADUATION THESIS

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DEPARTMENT OF INFORMATION TECHNOLOGY ENGINEERING

We certify that we have examined, and recommend to the University Steering Committee for Bachelor of engineering Graduation thesis entitled: **"CLASSIFICATION AND PESTICIDE RECOMMENDATION FOR MAIZE LEAF DISEASE BASED ON VGG16 MODEL"** submitted by **PHYO PA PA LWIN**, Roll No. **VI-IT 10 (NOVEMBER, 2024)** to the Department of Information Technology Engineering as the requirement for the Bachelor of Engineering degree **BE(IT)**.

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ABSTRACT

Maize is one of the most widely cultivated and economically important cereal crops in the world. However, maize production is often negatively affected by various leaf diseases such as Gray Leaf Spot, Common Rust, Fall Armyworm and Northern Leaf Blight. These diseases, if not detected and treated early, can lead to significant crop damage and yield loss. Traditional methods of disease detection rely on manual observation by agricultural experts or farmers, which are time-consuming, prone to error, and often unavailable in rural farming communities. Additionally, identifying and applying the correct pesticide requires specialized knowledge, and misuse can result in increased costs, environmental impact, and crop resistance.

This project proposes an intelligent system for the automated classification of maize leaf diseases and recommendation of appropriate pesticides using the VGG16 deep learning model, a type of Convolutional Neural Network (CNN). The model is trained on a labeled image dataset of diseased and healthy maize leaves to learn features and patterns that can accurately identify disease types. The methodology involves preprocessing the image dataset, splitting it into 80% training and 20% testing sets, and fine-tuning the VGG16 model to optimize classification performance. The system's performance is evaluated using metrics such as accuracy, precision, recall, and F1-score. In addition, a rule-based mapping system is implemented to connect each classified disease with its recommended pesticide, forming an end-to-end intelligent solution for farmers and agricultural researchers.

Experimental results show that the proposed system can achieve high accuracy in disease classification and provide relevant pesticide recommendations. This integration of deep learning and agricultural domain knowledge offers a promising tool for smart farming, enabling early intervention, reduced crop loss, and more sustainable pesticide usage. This study demonstrates the potential of artificial intelligence in revolutionizing traditional agriculture practices and contributes to building an accessible, automated, and practical solution for real-world crop disease management.

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